

Part B: Empirical Model and Analysis

Before starting, implement packages relevant for better regression outputs and graphs. Installing packages this way makes it easier to compile the RMarkdown document.

```
if(!require("ggplot2")) {install.packages("ggplot2"); library("ggplot2")}
library("ggplot2")
if(!require("stargazer")) {install.packages("stargazer"); library("stargazer")}
library("stargazer")
if(!require("coefplot")) {install.packages("coefplot"); library("coefplot")}
library("coefplot")
if(!require("ggeffects")) {install.packages("ggeffects"); library("ggeffects")}
library("ggeffects")
if(!require("car")) {install.packages("car"); library("car")}
library("car")
if(!require("rio")) {install.packages("rio"); library("rio")}
library("rio")
if(!require("rio")) {install.packages("rio"); library("rio")}
library("rio")
if(!require("Rmisc")) {install.packages("Rmisc"); library("Rmisc")}
library("Rmisc")
if(!require("knitr")) {install.packages("knitr"); library("knitr")}
library("knitr")
if(!require("kableExtra")) {install.packages("kableExtra"); library("kableExtra")}
library("kableExtra")
if(!require("patchwork")) {install.packages("patchwork"); library("patchwork")}
library("patchwork")
if(!require("lmtest")) {install.packages("lmtest"); library("lmtest")}
library("lmtest")
if(!require("tidyverse")) {install.packages("tidyverse"); library("tidyverse")}
library("tidyverse")
if(!require("jtools")) {install.packages("jtools"); library("jtools")}
library("jtools")
```

The analysis uses data from the World Bank. All variables are taken from the World Development Indicators (WDI). They can be accessed through this [LINK](#). The dataset includes 11 variables and uses data from 2016. I chose data from 2016 as it is the most complete with regards to the dependent variable (DV) I will investigate.

We first load the data, where NA values are defined to avoid problems where R does not recognize the variables as numerical:

```
library(readr)
dta <- read_csv("Desktop/EXAM - Analysis in R/THE_DATASET_E0.csv",
               na = c("NA", ".."))
```

Initial data investigation is crucial for raw data. Problems may occur at later stages of the analysis if the structure of the dataset is not investigated in the first step.

Since the variable titles the World Bank provides are long and overly detailed, I first rename the variables.

```
# renaming variables for better accessibility
fitting_names <- c(
  "country", "country_code", "elec_rural_perc", "gov_eff",
  "gdp_pc", "grid_loss", "elec_urban_perc", "elec_total",
  "rural_pop_share", "urban_pop_share", "pop_dens")

# Apply the names to original columns
colnames(dta) <- fitting_names
```

```
# checking names
names(dta)
```

```
## [1] "country"          "country_code"      "elec_rural_perc"  "gov_eff"
## [5] "gdp_pc"           "grid_loss"         "elec_urban_perc"  "elec_total"
## [9] "rural_pop_share"  "urban_pop_share"   "pop_dens"
```

I already log-transform the GDP per capita, adjusted for Purchasing Power Parity. This gives the values of GDP a “better fit” for the analysis and often is standard practice when including GDP as a variable. Doing this now makes subsetting the “finished and completed” dataset easier as it avoids unnecessary coding after subsetting the data in the next steps. I discuss the variable in more detail at a later point.

```
# Creating new variable for logGDP
dta$logGDP <- log(dta$gdp_pc)
```

With this variable included, we can now check the overall dimension of the dataset.

```
# dimension of dataset (number of observations and variables)
dim(dta)
```

```
## [1] 222 12
```

The preliminary analysis of the dataset shows that it still includes rows with copyright and other information.

```
# Using as.data.frame to remove additional tabble outpout
knitr::kable(as.data.frame(tail(dta[c(1,3,4)], 6)),
col.names = c("Country", "Rural Access %", "Gov. Eff."))
```

Country	Rural Access %	Gov. Eff.
Zimbabwe	21.9	-1.258878
	NA	NA
	NA	NA
	NA	NA
Data from database: World Development Indicators	NA	NA
Last Updated: 02/24/2026	NA	NA

The *dim* command still counts these rows as observations. In reality, the number of observations in this dataset is $222 - 5 = 217$. Since values in these rows are NA, R will exclude them when creating a subset in the next step. This observation is therefore just important when considering the *dim* command and will not be relevant in further applications.

In order to successfully complete the forthcoming regression analysis, we have to make sure the data is in numerical form: This presented a problem in preliminary testing of variables and regression models.

```
is.numeric(dta$elec_rural_perc)
```

```
## [1] TRUE
```

```
is.numeric(dta$gov_eff)
```

```
## [1] TRUE
```

Hypothesis

This paper investigates whether the quality of governance has an effect on the process electrification of rural areas in countries around the world. Electrification is a direct way a government can increase quality of life, access to education and the internet, as well as raising living standards (e.g. MacGinley, 2025). Additionally, “Individuals’ access to electricity is one of the most clear and un-distorted indication of a country’s energy poverty status” (World Bank). I therefore will investigate whether government efficiency has a significant effect on the electrification of rural areas in countries where rural electrification is not completed. The motivation behind the hypothesis is that giving access to electricity for rural populations is generally beneficial for those affected but rarely profitable. Building large grids and connecting remote, often small rural settlements to the central electricity grid therefore often requires specific state interventions and targeted policies. Electrification of rural areas through government programs, policies, and interventions offers a direct way to improve living standards of rural populations. Private companies often lack economic incentives and means to build an extensive, country-wide electricity grid. This hypothesis investigates whether the efficiency of governments and quality of governance is the central factor that determines the electrification of rural households. This insight, while also supported by other research (e.g. Gureye, Tadase & Ahmed, 2025), comes from personal experience during an internship in Egypt. The company, together with the Egyptian government, built grid connections for remote, rural Beduin settlements. People responsible told me that the lengthy process involved a lot of government cooperation and lobbying. This paper further investigates the potentially causal relationship between government effectiveness and electrification of rural areas in countries where rural electrification is not completed.

Dependent and Independent Variable

The dependent variable (DV) is rural electrification, describing the percentage of rural households that have access to some form of electricity. The variable is compiled using nationally representative household data. The complete specification can be found on the WORLD BANK DATABASE. The variable describes access to electricity of rural populations. Rural citizens live outside of urban hubs in small settlements or completely remote. We can check for the countries with the lowest and highest percentage of rural electrification and number of countries that have completed rural electrification completely (signaled by 100% rural electricity access):

```
dta$country[which.min(dta$elec_rural_perc)]
```

```
## [1] "Congo, Dem. Rep."
```

```
min(dta$elec_rural_perc, na.rm=TRUE)
```

```
## [1] 1
```

```
dta$country[which.max(dta$elec_rural_perc)]
```

```
## [1] "Andorra"
```

```
max(dta$elec_rural_perc, na.rm=TRUE)
```

```
## [1] 100
```

```
sum(dta$elec_rural_perc == 100, na.rm = TRUE)
```

```
## [1] 94
```

We see that the Democratic Republic of Congo has the lowest rural electrification number, with just 1% of rural areas that have access to electricity. Out of the 217 countries in the dataset, 94 have complete rural electricity access. For this analysis, the countries with 100% rural electricity access will be removed from the analysis to fit our hypothesis.

```
dta_low <- subset(dta ,elec_rural_perc < 100)
```

The subset now has different results for the highest and lowest rural electrification:

```
dta_low$country[which.min(dta_low$elec_rural_perc)]
```

```
## [1] "Congo, Dem. Rep."
```

```
min(dta_low$elec_rural_perc, na.rm=TRUE)
```

```
## [1] 1
```

```
dta_low$country[which.max(dta_low$elec_rural_perc)]
```

```
## [1] "Moldova"
```

```
max(dta_low$elec_rural_perc, na.rm=TRUE)
```

```
## [1] 99.9
```

Excluding countries with 100% rural electrification changes the structure of the data substantially. However, it is helpful for the purpose of the hypothesis. In these 94 excluded countries, rural electrification is already complete and cannot rise above 100%,

no matter how much the efficiency of a government changes. If the IV was significant with a positive coefficient (as we expect based on the hypothesis), it would suggest that, *ceteris paribus*, an increase in the Independent Variable (IV) leads to rural electrification that exceeds 100%, which is not possible in reality. In these countries, government efficiency might rise or fall significantly without affecting rural electrification in the model.

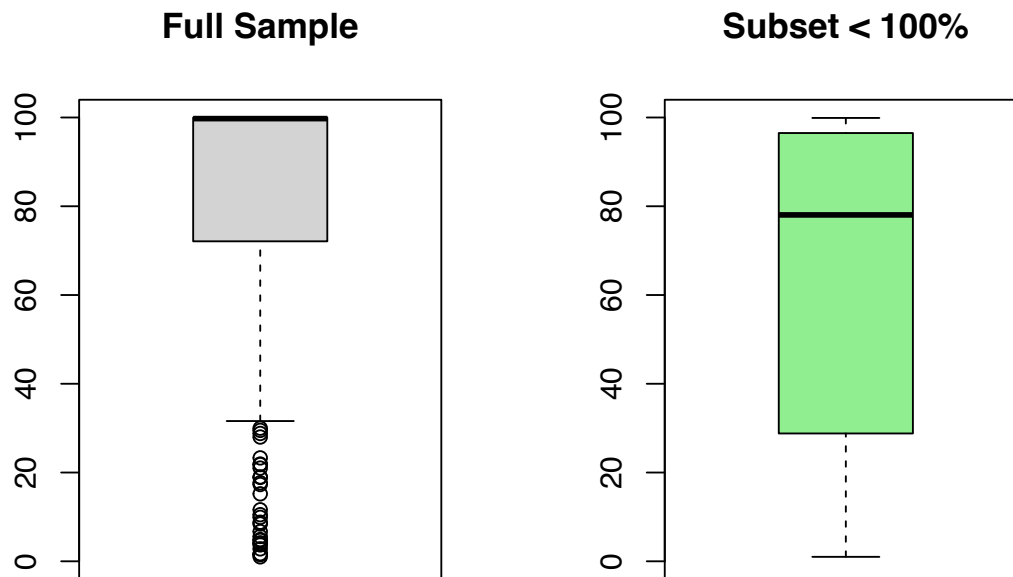
The reason for just removing countries with 100% rural electrification is the assumption that government efficiency is equally important to reach the last few rural areas without energy provisions. If 99% of rural areas are electrified, the 1% that is missing would be determined by the efficiency of the government to electrify even the most remote parts of a country. The subset focuses on countries that do not have complete rural electrification and (should) still work on expanding their energy grid and infrastructure. Only excluding countries with 100% rural electrification increases the number of observations, compared to subsetting for countries below e.g. 98%, which will be helpful when implementing control variables where some additional data might be missing.

Subsetting the data for rural electrification below 100% therefore helps with analysing the relationship between government efficiency and rural electrification where rural electrification is not complete. The following graphs presents the distribution of the full sample compared to the subset (< 100% rural electrification) to show how it changes the distribution:

```
par(mfrow = c(1, 2))

boxplot(dta$elec_rural_perc,
        main = "Full Sample",
        col = "lightgrey",
        ylim = c(0 , 100))

boxplot(dta_low$elec_rural_perc,
        main = "Subset < 100%",
        col = "lightgreen",
        ylim = c(0 , 100))
```



We can see that the initial model is very “top heavy”, given that a substantial number of countries (94 out of 217) have full rural electrification. The full sample has a median of close to 100%, while the median of the subset is about 78% rural electrification. The *summary* command again shows the difference that we see in the boxplot.

```
summary(dta$elec_rural_perc)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      1.00   73.10   99.75   80.11  100.00  100.00      18
```

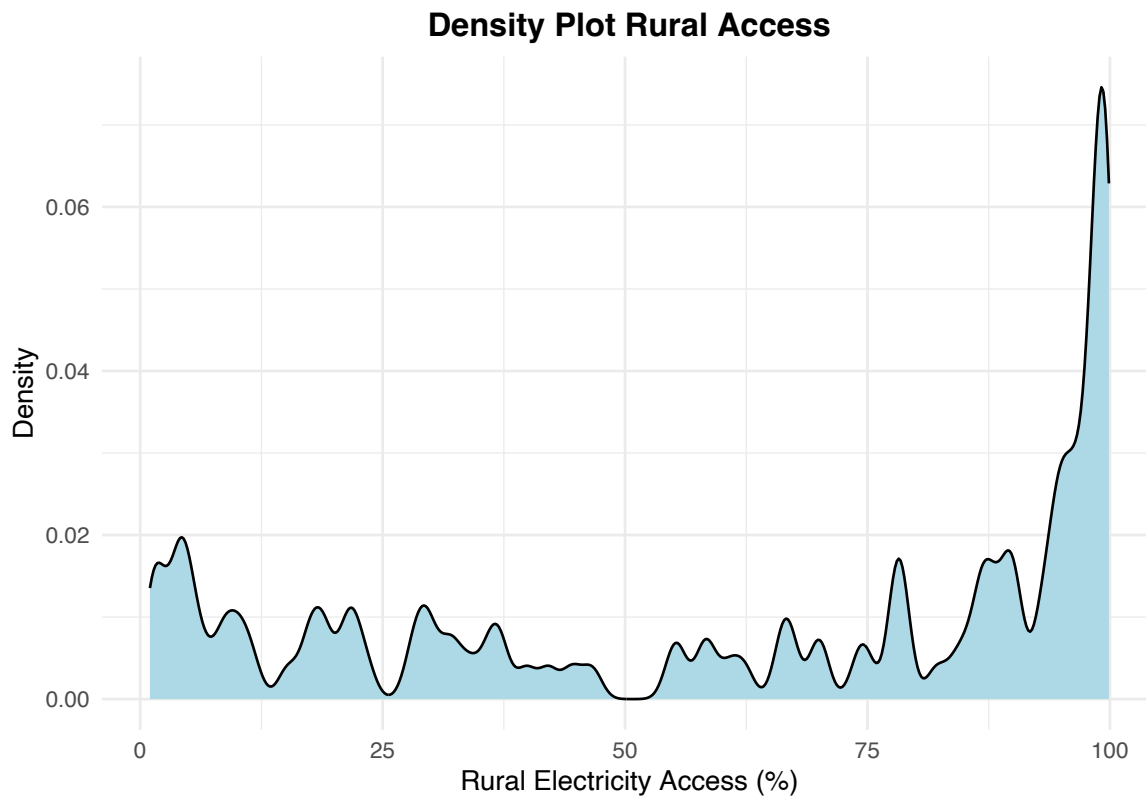
```
summary(dta_low$elec_rural_perc)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1.00   28.98   78.05   63.11   96.47   99.90
```

All regressions will use the subset with rural electrification below 100% for analysis. The comparison is exclusively used to present the large influence the subsetting has on the “original” variables. The density plot give additional insight into the distribution density.

```
ggplot(data = dta_low, mapping = aes(x = elec_rural_perc)) +
  geom_density(bw = 1, fill = "lightblue", alpha = 1) +
```

```
theme_minimal() +
xlab("Rural Electricity Access (%)") + ylab("Density") +
ggtitle("Distribution of Rural Electricity Access") +
theme(plot.title = element_text(hjust = 0.5,
face = "bold")) + ggtitle("Density Plot Rural Access")
```



The central independent variable (IV) for the model is government effectiveness. The variable describes government effectiveness on a scale from -2.5 to +2.5. The values are constructed using 30 different values that describe some aspect regarding the quality of governance. Government Effectiveness captures perceptions and views of the quality of public services, the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government's commitment to such policies. These aspects are evaluated by citizens, entrepreneurs, and experts in the public, private and NGO sectors from around the world. The full specification of the score and methodology can be found in the World Bank Database. I first check the *summary* of the variable.

```
summary(dta$gov_eff)
```

```
##      Min.   1st Qu.   Median     Mean  3rd Qu.     Max.      NA's
## -2.33689 -0.73750 -0.09601 -0.03260  0.68243  2.21462      17
```

```
summary(dta_low$gov_eff)
```

```
##      Min. 1st Qu.  Median     Mean 3rd Qu.     Max.
## -2.3369 -1.0002 -0.6341 -0.5978 -0.0752  1.0190
```

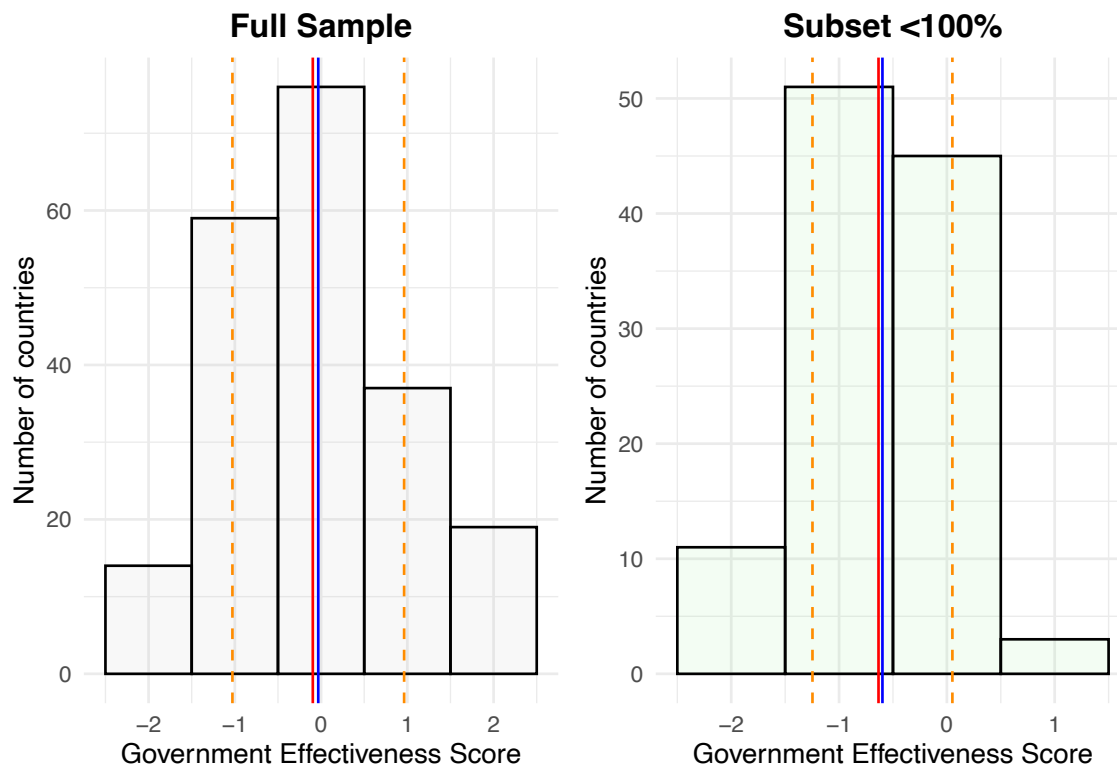
These graphs show the number of countries with efficiency scores in the full dataset and the subset with rural electrification below 100%.

```
p1 <- ggplot(data = dta, mapping = aes(x = gov_eff)) +
  geom_histogram(binwidth = 1, fill = "grey",
                 col = "black", alpha = 0.1) +
  theme_minimal() +
  ggtitle("Full Sample") +
  xlab("Government Effectiveness Score") + ylab("Number of countries") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold")) +
  geom_vline(xintercept = median(dta$gov_eff, na.rm=TRUE), col = "red") +
  geom_vline(xintercept = mean(dta$gov_eff, na.rm=TRUE), col = "blue") +
  geom_vline(xintercept = mean(dta$gov_eff, na.rm=TRUE) +
             sd(dta$gov_eff, na.rm=TRUE), col = "darkorange", lty = 2) +
  geom_vline(xintercept = mean(dta$gov_eff, na.rm=TRUE) -
             sd(dta$gov_eff, na.rm=TRUE), col = "darkorange", lty = 2)

p2 <- ggplot(data = dta_low, mapping = aes(x = gov_eff)) +
  geom_histogram(binwidth = 1, fill = "lightgreen",
                 col = "black", alpha = 0.1) +
  theme_minimal() +
  ggtitle("Subset <100%") +
  xlab("Government Effectiveness Score") + ylab("Number of countries") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold")) +
  geom_vline(xintercept = median(dta_low$gov_eff, na.rm=TRUE), col = "red") +
  geom_vline(xintercept = mean(dta_low$gov_eff, na.rm=TRUE), col = "blue") +
  geom_vline(xintercept = mean(dta_low$gov_eff, na.rm=TRUE) +
             sd(dta_low$gov_eff, na.rm=TRUE), col = "darkorange", lty = 2) +
```

```
geom_vline(xintercept = mean(dta_low$gov_eff, na.rm=TRUE)-
          sd(dta_low$gov_eff, na.rm=TRUE),
          col = "darkorange", lty = 2)
```

p1 + p2



The red lines show the median of the rural electrification, indicating the middle value in the dataset where half of the observations are above the median and half are below the median. The mean presents the average government effectiveness score from all observations in the dataset. The orange lines indicate the mean plus and minus one standard deviation. Most observations fall within that range.

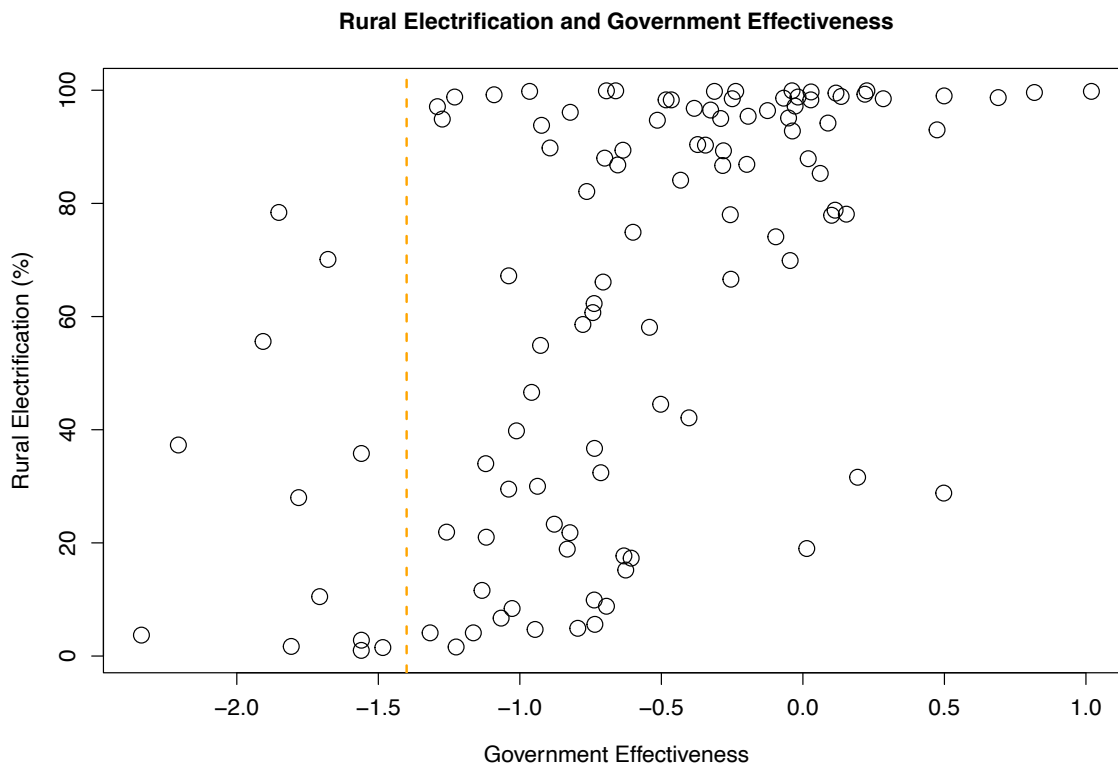
We now turn to the correlation between the variables of interest, rural electrification (in %) and government efficiency.

```
cor(dta_low$elec_rural_perc,dta_low$gov_eff)
```

```
## [1] 0.5632331
```

We can plot the correlation in a simple plot.

```
plot(jitter(dta_low$gov_eff, factor = 2), dta_low$elec_rural_perc,
     ylab = "Rural Electrification (%)",
     xlab = "Government Effectiveness",
     main = "Rural Electrification and Government Effectiveness",
     cex = 1.8,
     cex.lab = 1.2,
     cex.axis = 1.2,)
abline(v = -1.4, col = "orange", lwd = 2, lty = 2)
```



The correlation between Rural Electrification (in %) and Government Efficiency is about 0.56. The correlation between the two variables suggests a positive relationship meaning that, on average, an increase in government efficiency leads to an increase in rural electrification. It does *not* give a quantifiable effect as a regression coefficient would. The correlation does *not* give any information regarding a potentially causal relationship between the two variables either. A relatively high positive correlation is just a preliminary confirmation that the hypothesis may yield relevant results. The plotted values confirm this positive trend. Another thing this graph visualizes is that there are no countries with a government score below -1.4 that have complete rural electrification. While this is not directly relevant for the correlation between the

variables, it still is an interesting finding made available by this graph.

Bivariate Model and Analysis

After creating the subset for countries that do not have complete rural electrification, we can estimate the bivariate model. We regress rural electrification on

```
model_1 <- lm(elec_rural_perc ~ gov_eff, data = dta_low)
stargazer(model_1, report = "vcsp*")

# Output in R Console
stargazer(model_1, report = "vcsp*", type = "text")
```

Table 2: First Bivariate Model

	<i>Dependent variable:</i>
	elec_rural_perc
gov_eff	31.265 (4.414) p = 0.000***
Constant	81.798 (3.884) p = 0.000***
Observations	110
R ²	0.317
Adjusted R ²	0.311
Residual Std. Error	29.896 (df = 108)
F Statistic	50.180*** (df = 1; 108)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

The results show that government efficiency has a significant impact on electrification of rural areas. The positive and significant coefficient suggests that an increase in government efficiency leads to, *ceteris paribus*, an approximately 31.27% increase in rural electrification. The comparatively high coefficient is partly due to the small scale that government efficiency is measured on. An increase by 1 point describes a large difference in government efficiency. A 1 point increase on the scale can therefore be considered as a major governmental overhaul or large structural change of government institutions and functions. We instead look at a 0.1 point increase in government efficiency, which is more realistic than a 1 point increase. This step offers a more intuitive and accurate description of the effect induced by government efficiency. An increase in government efficiency by 0.1 points, *ceteris paribus*, leads to an increase in rural electrification of about 3.13%. The results are significant, signaled by the p-value below 0.01. We see that the R^2 value is 0.311, meaning that the model explains roughly 31.1% of the variation of the DV. This means that government efficiency alone explains about a third of the variation of rural electrification share. This is already a promising result, although 68.9% of the variance of the DV are not explained. In the following, I introduce additional control variables to see which other factors might influence rural electrification share.

Multivariate Model Analysis

First Control Variable

I include transmission losses as a share of total electricity output as an additional control variable. It is used as a proxy for electricity infrastructure quality and grid transmission effectiveness. It describes the share of energy that is lost in energy transmission due to technical constraints and grid ineffectiveness. The variable and complete specification is available in the WORLD BANK DATABASE. When the existing grid is inefficient (signaled by high transmission losses), grid extension to rural areas becomes more difficult. Governments that fail to provide efficient grids with high transmission rates might be equally unable to provide rural electricity access. Transmission losses lower monetary profits of energy distribution. An existing grid that has low transmission losses typically generates higher revenues as more of the energy that is produced actually reaches the consumers. Transmission losses are in part determined by continuous government efforts to modernize and sustain an efficient grid infrastructure. It therefore serves as a confounder as it reflects government policies and interventions regarding the quality of the energy grid.

```

model_2 <- lm(elec_rural_perc ~ gov_eff + grid_loss, data = dta_low)
stargazer(model_2, report = "vcsp*")

# Output in R Console
stargazer(model_2, report = "vcsp*", type = "text")

```

Table 3: First Multivariate Model

	Dependent variable:
	elec_rural_perc
gov_eff	26.363 (5.514) p = 0.00001***
grid_loss	-0.393 (0.329) p = 0.237
Constant	88.676 (6.444) p = 0.000***
Observations	80
R ²	0.281
Adjusted R ²	0.262
Residual Std. Error	29.446 (df = 77)
F Statistic	15.041*** (df = 2; 77)
Note:	*p<0.1; **p<0.05; ***p<0.01

The regression output shows that *grid_loss* has no significant effect on rural electrification share. More importantly, including transmission losses as a control variable reduces the amount of variance of the DV that is explained by the model, signaled by lower P^2 values compared to the initial bivariate model (see more in full model comparison below).

Second Control Variable

I also include GDP per capita, normalized to purchasing power parity to analyse whether the economic activity and wealth has a significant effect on rural electrification. It must be noted that GDP itself is by some considered to be an inaccurate measurement

of economic activity and wealth of a nation. It does not include any externalities produced by economic activities and informal markets within the economy. However, it remains a well documented and available variable that will be used here as a proxy of economic activity and wealth of a country, while acknowledging potential weaknesses of the variable. The contextual reason for including logGDP is that richer countries tend to have the financial capacity to build and finance energy grid extensions in rural areas. At the same time, several theories (e.g. Acemoglu & Robinson , 2012) argue that the wealth and prosperity of a nation is explained by the structure of their institutions. Therefore, GDP per Capita reasonably correlates with government efficiency. The following multivariate model estimates the effect of logGDP on rural electrification share.

```
model_3 <- lm(elec_rural_perc ~ gov_eff + logGDP, data = dta_low)
stargazer(model_3, report = "vcsp*")
```

Table 4: Second Multivariate Model

	<i>Dependent variable:</i>
	elec_rural_perc
gov_eff	9.988 (5.768) p = 0.087*
logGDP	24.787 (3.983) p = 0.000***
Constant	-146.624 (36.947) p = 0.0002***
Observations	104
R ²	0.523
Adjusted R ²	0.514
Residual Std. Error	25.167 (df = 101)
F Statistic	55.445*** (df = 2; 101)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

Controlling for GDP per capita with purchasing power parity shows that GDP is even more statistically significant than government efficiency. This makes sense, as wealthy

countries tend to have more resources to spend on grid infrastructure and electrification. An increase of GDP by 1%, ceteris paribus, leads to an increase in rural electrification by approximately 0.248% (level-log interpretation).

In the following I compare the bivariate model and the multivariate models.

```
stargazer(model_1,model_2,model_3,
  dep.var.labels = "Rural Electrification (%)",
  covariate.labels = c("Government Efficiency","Grid Losses",
    "log GDP per Capita"))

# Again add ,type = "text", for R Console Output
```

Table 5: Model Comparison

	<i>Dependent variable:</i>		
	Rural Electrification (%)		
	(1)	(2)	(3)
Government Efficiency	31.265*** (4.414)	26.363*** (5.514)	9.988* (5.768)
Grid Losses		−0.393 (0.329)	
log GDP per Capita			24.787*** (3.983)
Constant	81.798*** (3.884)	88.676*** (6.444)	−146.624*** (36.947)
Observations	110	80	104
R ²	0.317	0.281	0.523
Adjusted R ²	0.311	0.262	0.514
Residual Std. Error	29.896 (df = 108)	29.446 (df = 77)	25.167 (df = 101)
F Statistic	50.180*** (df = 1; 108)	15.041*** (df = 2; 77)	55.445*** (df = 2; 101)

Note: *(p<0.1); **(p<0.05); ***(p<0.01)

The table shows the differences between the bivariate model and multivariate model. In the first multivariate Model (2), Grid Loss is introduced as a control variable. The coefficient is not significant, despite its theoretical justification, showing that transmission losses do not affect rural electrification share. Additionally, including this control variable lowers R² and adjusted R² scores which is not beneficial for

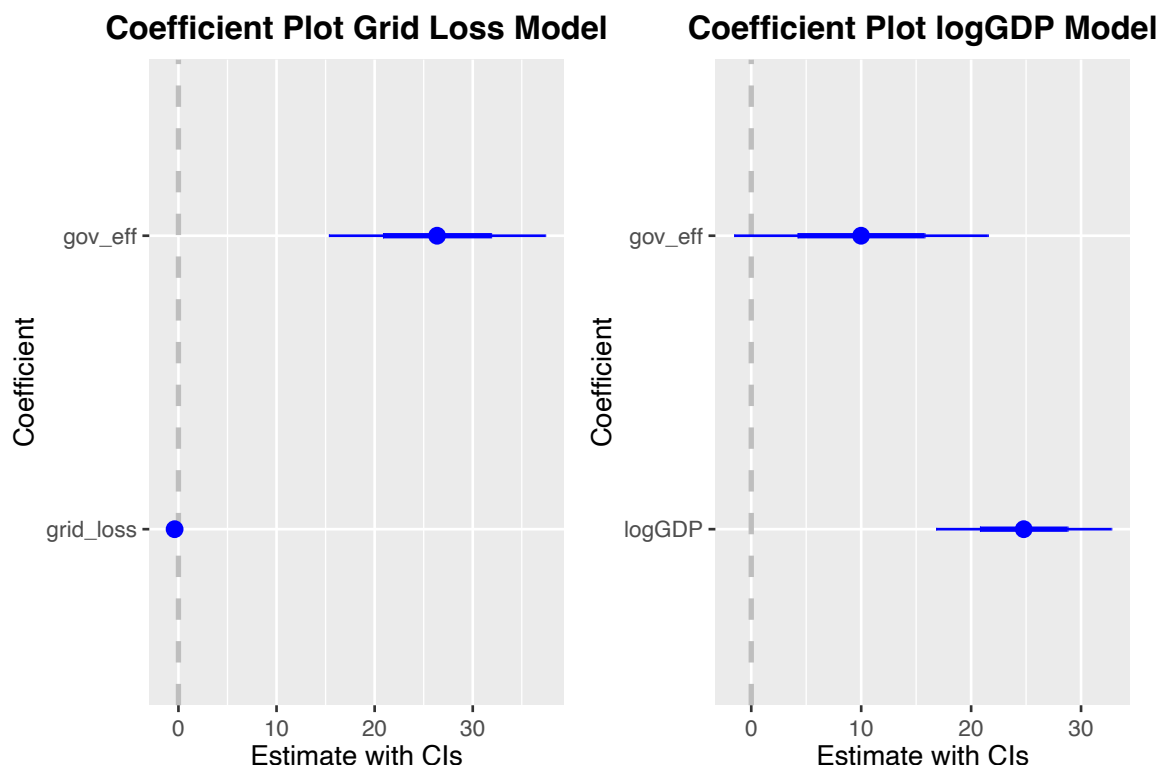
the explanatory power of the model regarding the variance of the DV. What is most interesting here is that the inclusion of logGDP in the second multivariate model (3) changes the coefficients significantly. The coefficient for government effectiveness is only a third of what it was in the bivariate model. Additionally, government effectiveness is only significant on the 10% level ($p < 0.1$), compared to significance on the 1% level ($p < 0.01$) in the bivariate model. However, the adjusted R^2 increases significantly and explains about 51.4% of the variation of the DV, compared to just 31.1% in the initial model. Note that the adjusted R^2 is a slightly better indicator compared to standard R^2 for multivariate models because it penalizes for the number of added variables. Additionally, the number of observations for the grid loss model drops by 30, suggesting that a lot of data is unavailable in the countries of interest. The result suggests that the effect of GDP per capita in terms of Purchasing Power Parity remains as the most significant variable in the model explaining electrification of rural areas

In the next step, I visualize the multivariate models further to show coefficient estimates and confidence intervals.

```
c2 <- coefplot(model_2,
  intercept = FALSE,
  title = "Coefficient Plot Grid Loss Model",
  xlab = "Estimate with CIs",
  ylab = "Coefficient",
  decreasing = TRUE,
  col = "blue") + theme(plot.title =
element_text(hjust = 0.5, face = "bold"))

c3 <- coefplot(model_3,
  intercept = FALSE,
  title = "Coefficient Plot logGDP Model",
  xlab = "Estimate with CIs",
  ylab = "Coefficient",
  decreasing = TRUE,
  col = "blue") + theme(plot.title =
element_text(hjust = 0.5, face = "bold"))

c2 + c3
```

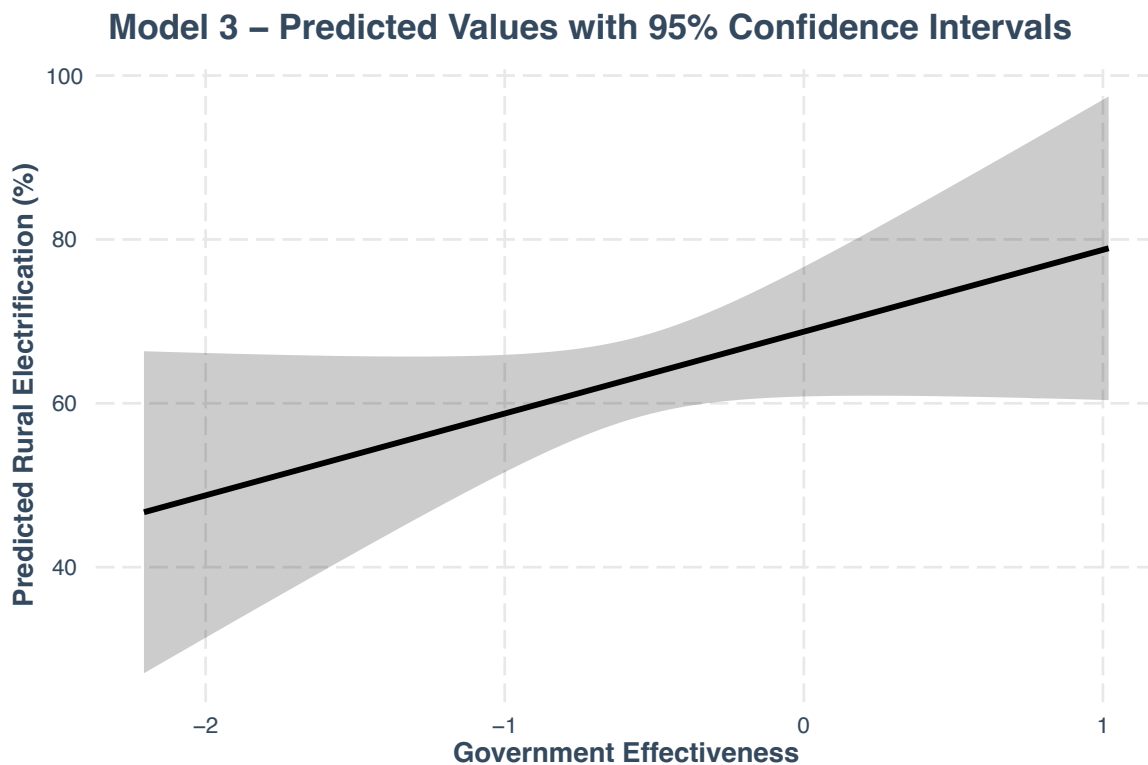


The coefficient plots match the regression outputs of the multivariate models. The confidence intervals for government effectiveness in the first bivariate model do not cross the zero line, confirming statistical significance ($p < 0.01$). The grid loss variable describing transmission losses shows to be insignificant, with a slightly negative coefficient and confidence intervals touching the zero line. The confidence intervals are barely visible as the predicted effect is marginal and rural electrification share barely changes due to grid losses. The plot confirms that the variable is statistically insignificant in the model. For the second bivariate model, the log-transformed GDP per capita (PPP adjusted) has an estimate of about 24.79, which is shown in the coefficient plot. Again, the confidence levels do not touch the zero line, confirming statistical significance ($p < 0.01$). However, the government effectiveness coefficient of approximately 9.99 shows confidence intervals that barely touch the zero line at 10% confidence level. This confirms that the effect and significance of government efficiency has become smaller as logGDP is introduced into the model.

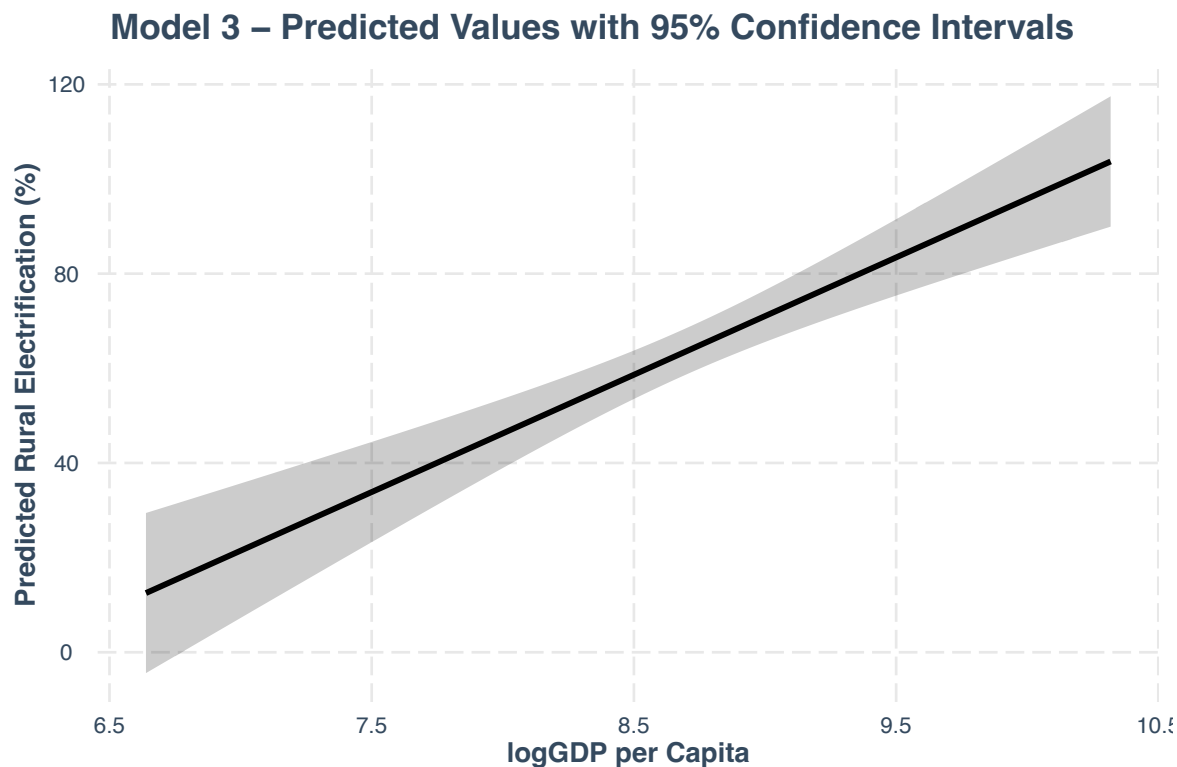
To plot predicted values, we use *model_3* which includes *logGDP per capita* and *government effectiveness* as the independent variables. The methodology comes from CRAN repository

```
# Using effect_plot
```

```
effect_plot(model_3,
             pred      = gov_eff,
             interval   = TRUE,
             x.label    = "Government Effectiveness",
             y.label    = "Predicted Rural Electrification (%)",
             main.title = "Model 3 - Predicted Values with 95% Confidence Intervals")
```



```
effect_plot(model_3,
             pred      = logGDP,
             interval   = TRUE,
             int.width  = 0.95,
             x.label    = "logGDP per Capita",
             y.label    = "Predicted Rural Electrification (%)",
             main.title = "Model 3 - Predicted Values with 95% Confidence Intervals")
```



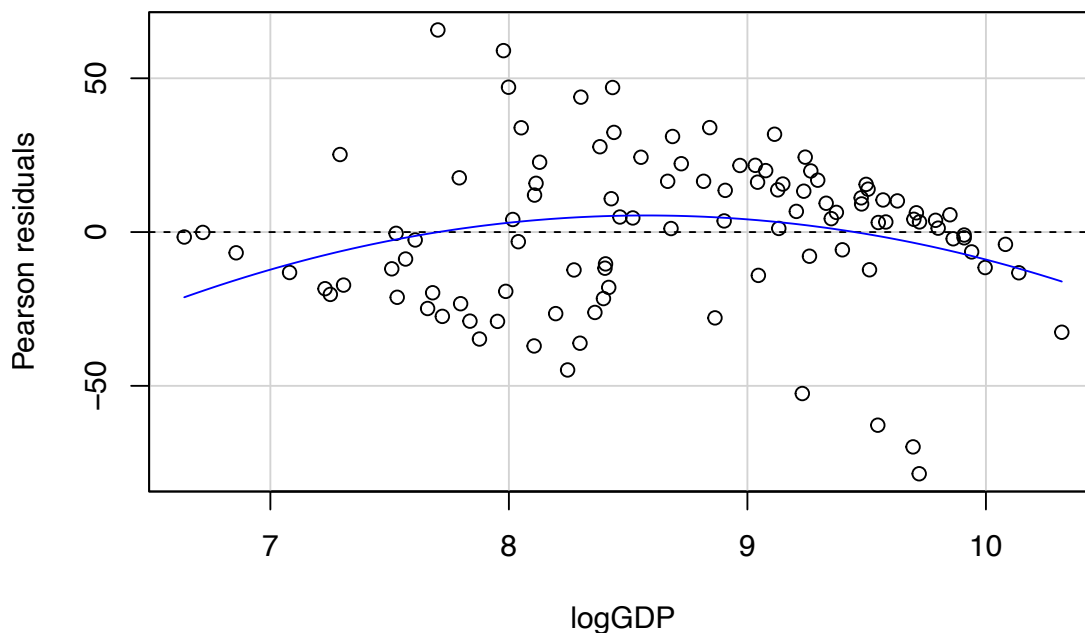
In the initial bivariate model, government efficiency has a significant impact on rural electrification in countries where rural electrification is not complete. When introducing GDP per capita as a control variable, the significance and estimated effect of government efficiency shrinks. The initial hypothesis might not hold under scrutiny and during more extensive hypothesis testing. The results and estimated effects of government efficiency may become insignificant when introducing additional control variables whose inclusion is contextually and empirically warranted. Using GDP per capita and government efficiency as IVs reveals that GDP per capita help explain a large part of the variation of rural electricity access. We may therefore say, based on available dataset from 2016, that rural electricity access, in countries where it is below 100%, is in part determined by the effectiveness of a government to provide effective governance. The analysis reveals additional information about what determines rural electricity access. Firstly, it seems that GDP per capita has a significant impact on the provision of energy infrastructure for rural areas. It does not seem to be the case that the quality of the grid and transmission losses hinder grid extensions in rural areas. The plot showing predicted values with 95% confidence intervals supports the hypothesis, showing that the effects are not random.

Testing Regression Assumptions

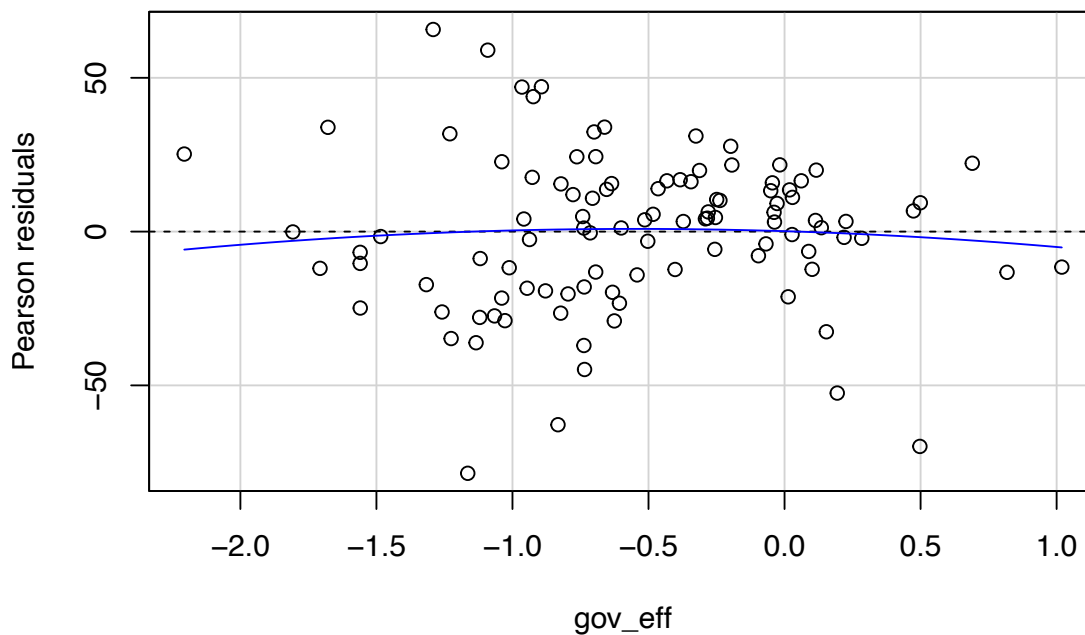
I test regression assumptions for the multivariate *model_3* including *logGDP* and *government efficiency* variables, as it seems to be the most “promising” model considering R^2 values and significance of the variables.

The first regression assumption we test is the Zero Conditional Mean assumption. Satisfying this assumption is crucial as otherwise the estimated coefficients and OLS results are biased. The Zero Conditional Mean assumption can be tested using residual plots.

```
residualPlot(model_3, variable = "logGDP")
```



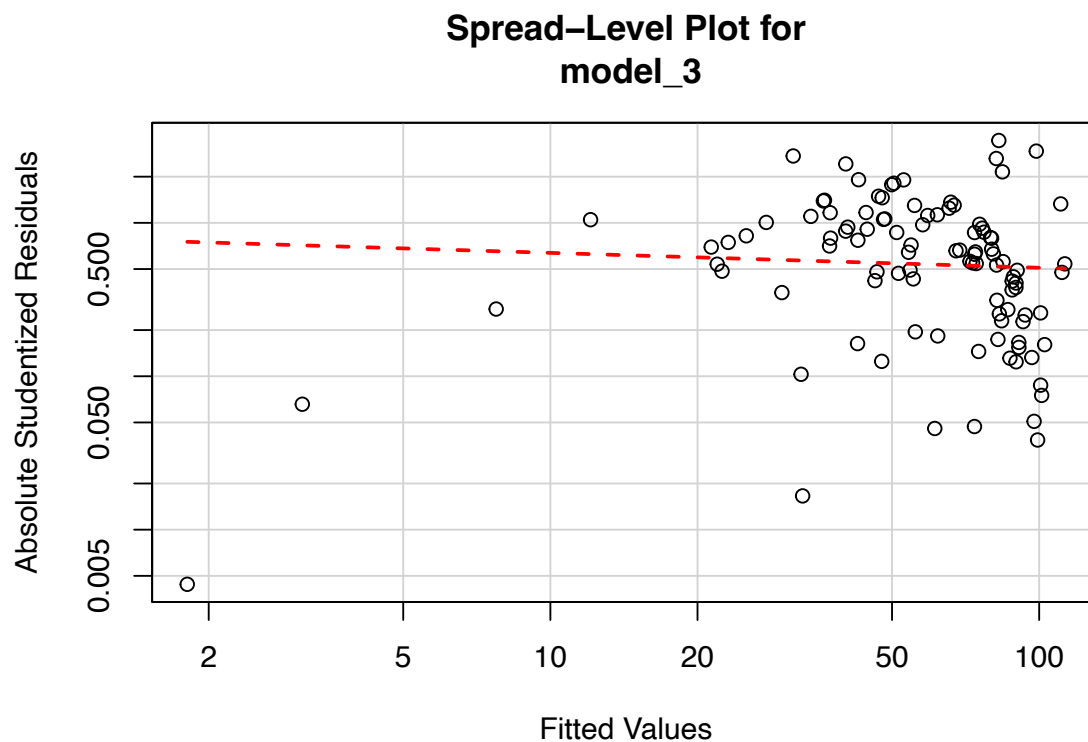
```
residualPlot(model_3, variable = "gov_eff")
```



It should be noted that the results of this analysis do not conclude that the Zero Conditional Mean assumption is violated in reality. The results merely give an indication whether the assumption is violated, not an exhaustive answer. In this case, the plot shows a slightly curved line for both variables. This is not overly concerning and does not pose an immediate problem for the model. However, one should be aware of potential violations and implement more rigorous tests before coming to a final conclusion.

The second assumption I will test regarding *model_3* is homoskedasticity. It means that the “error term has the same variance given any value of the explanatory variable.” (Script 7). The opposite would be heteroskedasticity

```
spreadLevelPlot(model_3, col.lines = "red")
```



```
##
## Suggested power transformation:  1.097046
```

```
bptest(model_3)
```

```
##
## studentized Breusch-Pagan test
##
## data:  model_3
## BP = 5.5542, df = 2, p-value = 0.06222
```

We see that residuals are not evenly scattered across the plot and the line is not perfectly horizontal. The p-value (0.06) of the Breusch-Pagan test confirms this. This suggests a deviation from perfect homoskedasticity which is not overly concerning. Homoskedasticity is not necessary to get unbiased results. If we still were to implement robust standard errors, the results would look like this.

```
# Creating robust standard errors
robust_mod_3 <- hccm(model_3, type="hc1")
mod_3_robust <- coeftest(model_3, robust_mod_3)
```

```
#Displaying results
```

```
stargazer(model_3, mod_3_robust)
```

Table 6

	<i>Dependent variable:</i>	
	elec_rural_perc	
	<i>OLS</i>	<i>coefficient test</i>
	(1)	(2)
gov_eff	9.988* (5.768)	9.988 (7.369)
logGDP	24.787*** (3.983)	24.787*** (4.675)
Constant	-146.624*** (36.947)	-146.624*** (43.504)
Observations	104	
R ²	0.523	
Adjusted R ²	0.514	
Residual Std. Error	25.167 (df = 101)	
F Statistic	55.445*** (df = 2; 101)	

Note:

*p<0.1; **p<0.05; ***p<0.01

We see that when implementing robust standard errors, government efficiency loses significance at the 10% level ($p < 0.1$) and becomes insignificant in the model. However, given that the Breusch-Pagan test result is only significant at the 10% level, I decide to ignore the heteroskedasticity to not unnecessarily sacrifice efficiency of the model. We therefore accept a small level of

We lastly test an interaction term that suits our hypothesis. The hypothesis states that government intervention is necessary because private companies lack economic incentives to build rural energy grids. The previous models implicitly assume that government effectiveness has the same effect on rural electrification, regardless of how rural the country is. The interaction term (*government effectiveness * rural population share*) captures the difficulty of rural electrification. Government effectiveness is conditional on the size of rural populations, as countries with larger rural population shares

might require more extensive government interventions and policies to achieve rural electrification.

```
model_interact <- lm(elec_rural_perc ~ gov_eff +
                     logGDP + (gov_eff * rural_pop_share), data = dta_low)

stargazer(model_interact, report = "vcsp")
```

Table 7: Model with interaction term

	<i>Dependent variable:</i>
	elec_rural_perc
gov_eff	10.257 (14.051) p = 0.468
logGDP	26.104 (5.061) p = 0.00001***
rural_pop_share	0.062 (0.189) p = 0.744
gov_eff x rural_pop_share	-0.017 (0.241) p = 0.945
Constant	-161.583 (51.525) p = 0.003***
Observations	104
R ²	0.524
Adjusted R ²	0.505
Residual Std. Error	25.396 (df = 99)
F Statistic	27.270*** (df = 4; 99)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

The results from introducing an interaction term suggest that while the estimate of logGDP per capita remains significant, government effectiveness does not. We also see that including the interaction term does not improve (adjusted) R² substantially. The

interaction term, although contextually reasonable, does not show a significant effect on rural electrification share.

Brief conclusion

Given that no OLS assumptions were directly violated and government efficiency remains significant in the model when controlling for logGDP, we can conclude that the significant effect of government efficiency on rural electrification share remains. While this model does not suffice to establish a clear causal relationship, it is a valuable finding that sheds light on rural electrification process in countries where countrywide electrification is not complete. The findings suggest that an increase in government efficiency, signaled by more effective policies and government interventions increases rural electrification. Other variables, like the quality of the overall energy grid and the interaction term between government efficiency and rural population share do not yield significant results in this model. However, when including the interaction term, government effectiveness becomes insignificant in the model. Any conclusions drawn from this model must acknowledge the importance of GDP per capita and the fact that government effectiveness may become insignificant once more control variables are added.

1 Potential Improvements and Limitations

The hypothesis relies on government efficiency as the main independent variable. This may be a weakness, as government efficiency is no directly observable value. It is constructed using expert opinions to determine it in a range from -2.5 to + 2.5. It is a valuable variable that describes the overall function of a government and should not be discarded entirely. It is not clear how a country can achieve a change of 0.1 government efficiency. The quality of a given policy is equally hard to quantify. Other, more quantifiable indicators may be helpful here. These could include, but are not limited to a political stability index or rule of law index. Apart from using such indicator, it might be promising to find other proxies that describe the effectiveness government regarding energy more specifically. These could include public spending on energy infrastructure, average number of power outages, or energy prices and energy subsidies. Using panel data and advanced regression models seems promising for further evaluating the initial hypothesis. If such cases exist, it might be interesting to investigate whether a decrease in rural electrification over time is linked to a decrease in government effectiveness. Thirdly, the exact definitions of what counts as rural households is not

explicitly stated in the description of the variable. There is no defined limit, and it is unclear whether a small municipality with e.g. 150 households still counts as a rural settlement. This may be relevant for additional specifications of the model, like adding rural population share as a control variable. It also helps to understand the underlying challenges of rural energy provisions.

Regarding the choice of data, I would use a more extensive panel data with more available data points. This may greatly improve the missing variables in the dataset may contribute to a selection bias in the data and subsequent model. Countries with GDP per capita below the mean have more missing variables for rural electrification compared to the countries above the mean GDP, where no data is missing.

```
nrow(subset(dta, dta$gov_eff < mean(dta$gov_eff, na.rm = TRUE)))
```

```
## [1] 110
```

```
nrow(subset(dta, dta$gov_eff > mean(dta$gov_eff, na.rm = TRUE)))
```

```
## [1] 95
```

```
sum(is.na(subset(dta,
dta$gov_eff < mean(dta$gov_eff, na.rm = TRUE))$elec_rural_perc))
```

```
## [1] 11
```

```
sum(is.na(subset(dta,
dta$gov_eff > mean(dta$gov_eff, na.rm = TRUE))$elec_rural_perc))
```

```
## [1] 2
```

```
# Calculating the percentage of missing values below and above mean
```

```
11/110
```

```
## [1] 0.1
```

```
2/95
```

```
## [1] 0.02105263
```

For countries with available data for government efficiency below the mean of the original dataset, 11 observations regarding rural electrification are missing. For countries with government efficiency above the mean, only 2 observations are missing regarding rural electrification. This suggests that well governed countries tend to be over-represented in the dataset and weaken the model and following conclusion. For well governed countries, 2% of the data is missing while 10% of data for rural electrification is missing for countries with low government efficiency. Values might *not* be missing at random. Future models and datasets should therefore ideally include as many datapoints for each variables and all countries. The model and method presented in this example do give valuable insights but must always be considered in light of these limitations.

Bibliography

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Appendix

I briefly check for multicollinearity in response to the theory by Acemoglu & Robinson (2012), who claim that the quality and nature of government institutions determines the wealth and prosperity of a nation. Multicollinearity occurs when IVs are strongly correlated, leading to inflated standard errors and makes estimated coefficients more sensible to model changes. According to Acemoglu & Robinson (2012), multicollinearity between government efficiency and logGDP should be expected.

```
vif(model_2)
```

```
##   gov_eff grid_loss  
## 1.090828 1.090828
```

```
vif(model_3)
```

```
##   gov_eff   logGDP  
## 1.953044 1.953044
```

For either model, multicollinearity is overly concerning. Generally, VIF values between 1 and 5 suggest that there is “moderate, but acceptable correlation: no further action is needed” (Script 7). Since this is the case here, we can test for another regression assumption.